

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONE

Supplementary Winter Examination – 2024

Course: B. Tech **Branch:** Electrical Engineering/ Electrical Engineering (Electronics and Power)/ Electrical & Electronics Engg. / Electrical & Power Engineering
Subject Code & Name: (BTEEC702_Y20) High Voltage Engineering **Semester :** VII
Max Marks: 60 **Date:** 07/ 02/ 2025 **Duration:** 3 Hr.

Instructions to the Students

1. Each question carries 12 marks
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/ expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

		(Level/ CO)	Marks
Q. 1	Objective type questions. (Compulsory Question)		12
1	Which dielectric materials are used in high-voltage equipment to serve the dual purpose of insulation and heat dissipation? (a) Solids (b) Liquids (c) Air (d) Gas	(L1/ CO1)	1
2	Average electrical field is the magnitude of electrical field (a) at midpoint between conductors (b) ratio of potential difference to the distance between the conductors (c) at surface of the lower potential electrode (d) ratio of potential difference to half the distance between the conductors.	(L2/ CO2)	1
3	An electronegative gas is one in which (a) positive ions are formed along with electrons (b) gas has inherent negative charge (c) gas is ionized due to electron bombardment (d) electron gets attached to form negative ion	(L1/ CO2)	1
4	Ionization coefficients α , γ are functions of (a) applied voltage (b) pressure and temperature (c) electric field (d) ratio of electric field to pressure	(L2/ CO2)	1

51641377	5	The parameters that affect the breakdown strength of liquids is (a) hydrostatic pressure and temperature (b) dissolved impurities (c) dielectric constant (d) pressure, temperature, dissolved impurities and suspended particles.	(L4/ 002)	1	51641377
51641377	6	Paper insulation is mainly used in (a) cables and capacitors (b) transformers (c) rotating machines (d) circuit breakers	(L1/ 002)		51641377
51641377	7	Material that is used in surge arresters for BHV and UHV power systems is (a) silicon carbide (b) zinc oxide (c) aluminium oxide (d) metal oxides.	(L4/ 002)	1	51641377
51641377	8	Overhead transmission lines are protected from lightning overvoltages (a) by counter poise wires (b) by protector tubes (c) by ground or shield wires above the main conductors (d) by shunt reactors.	(L2/ 002)		51641377
51641377	9	Tesla coil is used for (a) generation of sinusoidal output voltages (b) generation of very high voltages (c) generation of rectangular voltages (d) generation of high frequency ac voltages	(L1/ 002)	1	51641377
51641377	10	A generating voltmeter is used to measure (a) impulse voltages (b) ac voltages (c) dc voltages (d) high-frequency ac voltages	(L1/ 002)		51641377

11	Fault location in an HV cable is done by (a) voltage withstand test (b) partial discharge scanning tests (c) life tests (d) impulse testing	(L4/ 003)	1
12	Most important tests conducted on isolators and circuit breakers are (a) voltage withstand tests (b) short circuit tests (c) high current tests (d) temperature rise tests.	(L4/ 004)	
Q. 2 Solve the following.			
A)	What are surge voltages and how are they distributed in the windings of a power apparatus like a transformer winding?	(L2/ 001)	6
B)	Define Townsend's first and second ionization coefficients. How is the condition for breakdown obtained in a Townsend discharge?	(L5/ 002)	6
Q.3 Solve the following.			
A)	Explain with neat sketch, the time lags for breakdown in gases.	(L4/ 002)	6
B)	What is Paschen's law? Write a short note on glow discharge and arc discharge.	(L2/ 003)	5
Q. 4 Solve Any Two of the following.			
A)	Explain the conduction and breakdown in pure liquids.	(L4/ 004)	6
B)	Explain the phenomenon 'treeing and tracking' in solid insulating materials under electrical stress. How does it lead to breakdown?	(L2/ 004)	6
C)	Explain the applications of insulating materials in power transformers.	(L2/ 003)	6
Q.5 Solve Any Two of the following.			
A)	Derive the expressions for the voltage and current waves on long transmission lines and obtain the surge impedance of the line.	(L5/ 005)	6
B)	What is meant by insulation co-ordination? How are the protective devices chosen for optimal insulation level in a power system?	(L4/ 004)	6

Q	Give the Marx circuit arrangement for multistage impulse generators. How is the basic arrangement modified to accommodate the wave time control resistances?	(L2/004)	6
Q. 6	Solve Any Two of the following.		12
A)	Describe with a neat sketch, the working of a Van de Graaff generator. What are the factors that limit the maximum voltage obtained?	(L3/004)	
B)	Explain capacitance voltage transformer (CVT) with a neat sketch. What are its advantages and disadvantages?	(L2/005)	
Q	Explain the method of impulse testing of high voltage transformers. What is the procedure adopted for locating the failure?	(L3/003)	6
*** End ***			